Cycles through 1 for the $qn + 1$ map: no cycles with three odd steps, and constraints on longer cycles

Trevor Hackman

Abstract

For the $qn + 1$ map with q an odd integer at least 3, we prove that 1 never lies on a cycle with exactly three distinct odd numbers. This settles the three-step case of a conjecture of Santos that 1 is periodic only for $q = 5$ and for Mersenne numbers $q = 2^e - 1$. We found that for every such q at least 2^{22003}, a lower bound and an upper bound cannot hold at the same time. Bauer and Bennett (2018) give an irrationality measure whose exponent, about 1.6149, lies just below the golden ratio, the threshold at which these bounds conflict. Every smaller q is ruled out by a direct computation.

1 Introduction

Fix an odd integer $q \geq 3$. For an odd positive integer n, let v be the largest integer such that 2^v divides $qn + 1$. The map C sends n to

$$C(n) = \frac{qn + 1}{2^v}.$$

The orbit of n is the sequence n, $C(n)$, $C(C(n))$, …. When $q = 3$, C is the odd-to-odd or accelerated form of the Collatz map, and the Collatz conjecture, posed by Lothar Collatz in 1937 and still open, asserts that every orbit eventually reaches 1. For $q = 3$ the number 1 is fixed, since $3 \cdot 1 + 1 = 4$. For larger q, heuristic arguments suggest that most orbits grow without bound and that cycles are rare, although no orbit has been proved to diverge.

We say that 1 is periodic if its orbit returns to 1, and ask for which q this happens. We classify the answers by the length ℓ of the cycle through 1, that is, the number of distinct odd numbers on it.

For $\ell = 1$ the cycle is $1 \to 1$, which happens exactly when $q + 1$ is a power of 2, that is $q = 2^e - 1$ (Santos, Proposition 15). For $\ell = 2$, the only q is 5, with $1 \to 3 \to 1$ (Santos, Proposition 16; a short proof also follows from Beukers's theorem on $x^2 + D = 2^n$, stated as Theorem 1.8 in Bauer and Bennett (2002)). Santos conjectured that these are the only cases: 1 is periodic only for $q = 5$ and $q = 2^e - 1$.

Theorem 1. No odd $q \geq 3$ has 1 on a cycle of C of length 3.

So the conjecture holds for every cycle through 1 of length at most 3. For longer cycles we prove an exact criterion for an effective answer, and finiteness.

Theorem 2 (golden-ratio criterion, informal). Fix $\ell \geq 2$, and suppose the roots of 2 that this length produces have an approximation exponent below the golden ratio. Then any q with 1 on a cycle of C of length ℓ lies below a bound we can compute. Checking every candidate up to that bound finds every such q, or shows that none exist. The precise statement is Theorem 2′ in Section 6.

Theorem 3. For each $\ell \geq 2$, only finitely many odd q have 1 on a cycle of C of length ℓ.

Theorem 3 is ineffective: it rests on Ridout's theorem and gives no bound on q. (A finiteness result is called effective if its proof gives an explicit bound on any exception, so that a finite computation can find them all, and ineffective if it gives no such bound.) Theorem 1 is effective and is completed by computation.

2 Definitions and notation

Throughout, $q \geq 3$ is an odd integer and C is the map defined in Section 1.

The 2-adic valuation. For a nonzero integer n, $v_2(n)$ is the largest integer v such that 2^v divides n.

Cycles through 1. An ℓ-odd cycle through 1 is a cycle of C of length ℓ containing 1, written

$$1 = x_0 \to x_1 \to \cdots \to x_{\ell-1} \to 1.$$

Its elements are odd and distinct.

Halvings. The halving counts $k_1, \ldots, k_\ell$ of the cycle are defined by

$$qx_{i-1} + 1 = 2^{k_i} x_i \qquad (1 \leq i \leq \ell,\ x_\ell = 1),$$

so $k_i = v_2(q \cdot x_{i-1} + 1)$. The total number of halvings is $h = k_1 + \cdots + k_\ell$, and $S_i = k_1 + \cdots + k_i$.

The window. On an ℓ-odd cycle through 1 with h halvings in all, write $2^{h/\ell} = q + \delta$, so δ is the fractional part of $2^{h/\ell}$. The window is a lower bound and an upper bound on δ.

Near misses. For an element x_i other than 1 on an ℓ-odd cycle through 1, set $n_i = ih - \ell S_i$. The near miss of x_i is $2^{n_i/\ell}$. The relative error ρ_i is defined by

$$x_i = 2^{n_i/\ell}(1 + \rho_i).$$

Approximation exponents. Let θ be irrational and let $\mathcal{D}$ be a set of positive integers. Then θ has effective approximation exponent μ with respect to $\mathcal{D}$ when there are explicit constants $c > 0$ and d_0 such that

$$\left|\theta - \frac{p}{d}\right| > c \cdot d^{-\mu}$$

for every integer p and every $d \in \mathcal{D}$ with $d \geq d_0$. Write $\mathcal{D}_\ell$ for the set of denominators 2^j and $\ell \cdot 2^j$, where j is a nonnegative integer.

The golden ratio. $\varphi = (1 + \sqrt{5})/2$.

The order of 2. For an odd positive integer m, the order of 2 modulo m is the smallest positive integer d such that $2^d \equiv 1 \pmod{m}$.

Other conventions. An implied constant in O, $\ll$, $\gg$, or $\asymp$ depends only on ℓ. In a heuristic remark, $\lesssim$ and $\gtrsim$ mean at most or at least roughly, ignoring constants.

Three odd steps. When $\ell = 3$, write the cycle as $1 \to x \to y \to 1$. The three steps are

$$q + 1 = 2^{k_1} x, \qquad qx + 1 = 2^{k_2} y, \qquad qy + 1 = 2^{k_3}.$$

3 The structure of a three-step cycle

Lemma 1. $k_3 \geq k_1 + 3$, and $y \equiv 1 \pmod{2^{k_1}}$.

Proof. The odd elements other than 1 are at least 3, so $x \geq 3$ and $y \geq 3$. The first step is $q + 1 = 2^{k_1}x$, and therefore $q + 1 \geq 3 \cdot 2^{k_1}$, that is $q \geq 3 \cdot 2^{k_1} - 1$. The third step is $2^{k_3} = qy + 1$, so

$$2^{k_3} \geq (3 \cdot 2^{k_1} - 1) \cdot 3 + 1 = 9 \cdot 2^{k_1} - 2.$$

Since $2^{k_1} \geq 2$, we have $9 \cdot 2^{k_1} - 2 \geq 8 \cdot 2^{k_1} = 2^{k_1+3}$, and hence $k_3 \geq k_1 + 3$. From $q + 1 = 2^{k_1}x$ we also have $q \equiv -1 \pmod{2^{k_1}}$. Thus $qy + 1 \equiv 1 - y \pmod{2^{k_1}}$. But $qy + 1 = 2^{k_3}$ is divisible by 2^{k_1}, because $k_3 \geq k_1$, so $1 - y \equiv 0 \pmod{2^{k_1}}$, that is $y \equiv 1 \pmod{2^{k_1}}$. □

Lemma 2 (cycle equation).

$$2^h = q^3 + q^2 + 2^{k_1}q + 2^{k_1+k_2}.$$

Proof. Since $h = k_1 + k_2 + k_3$,

$$2^h = 2^{k_1+k_2} \cdot 2^{k_3}.$$

The third step is $qy + 1 = 2^{k_3}$, so

$$2^h = 2^{k_1+k_2}(qy + 1) = 2^{k_1}(q \cdot 2^{k_2}y + 2^{k_2}).$$

The second step is $qx + 1 = 2^{k_2}y$, and therefore

$$2^h = 2^{k_1}(q(qx + 1) + 2^{k_2}) = 2^{k_1}(q^2x + q + 2^{k_2}) = q^2(2^{k_1}x) + 2^{k_1}q + 2^{k_1+k_2}.$$

The first step is $q + 1 = 2^{k_1}x$, so

$$2^h = q^2(q + 1) + 2^{k_1}q + 2^{k_1+k_2} = q^3 + q^2 + 2^{k_1}q + 2^{k_1+k_2}.$$

□

Lemma 3 (cube window). $q^3 < 2^h < (q + 1)^3$. Hence 3 does not divide h and $q = \lfloor 2^{h/3} \rfloor$.

Proof. The lower bound is immediate from Lemma 2. Since $2^{k_1+k_2} = (q(q + 1) + 2^{k_1})/y \leq (q + 1)(3q + 1)/9$ by $y \geq 3$, and $2^{k_1} \leq (q + 1)/3$,

$$2^h - q^3 \leq q^2 + \frac{q(q + 1)}{3} + \frac{(q + 1)(3q + 1)}{9} = \frac{15q^2 + 7q + 1}{9} < (q + 1)^3 - q^3.$$

A power of 2 strictly between consecutive cubes is not a cube, so $3 \nmid h$. □

Lemma 4 (2-adic structure of x). Let $t = v_2(x - 1)$ and $s = (x - 1)/2^t$. Then either $t = 2k_1$, or $t < k_1$ and $k_1 = t + v_2(s + 1)$, or $t = k_1$.

Proof. Suppose $t > k_1$. Then $qx + 1 = 2^{k_1}x^2 - (x-1)$ has 2-adic valuation k_1, so $k_2 = k_1$ and $y = x^2 - w$ with $w = (x-1)/2^{k_1}$ even. Substituting $x = 1 + 2^{k_1}w$,

$$qy + 1 = 2^{k_1} + w(1 - 3 \cdot 2^{k_1}) + 2^{2k_1}w(3 - 2w) + 3 \cdot 2^{3k_1}w^2 + 2^{4k_1}w^3.$$

The last three terms are divisible by $2^{2k_1+v_2(w)}$ and $1 - 3 \cdot 2^{k_1}$ is odd, so if $v_2(w) \neq k_1$ then $v_2(qy+1) = \min(k_1, v_2(w)) \leq k_1$, contradicting Lemma 1. Hence $v_2(w) = k_1$, that is $t = 2k_1$.

Suppose $t < k_1$. Then $2^{k_1}x^2 - x + 1 = 2^t(2^{k_1-t}x^2 - s)$ with the bracket odd, so $k_2 = t$, $y = 2^{k_1-t}x^2 - s$, and

$$qy + 1 = (2^{k_1} - 1)\,2^{k_1-t}x^2 + (1+s) - 2^{k_1}s + 2^{k_1+t}sy.$$

The last two terms are divisible by 2^{k_1} and the first has valuation exactly $k_1 - t$, so $v_2(s+1) \neq k_1 - t$ would give $v_2(qy+1) < k_1$, again contradicting Lemma 1. □

Lemma 5. $2^{k_1} \leq 2(x-1)$ and $x > \sqrt{q/2}$.

Proof. In the three cases of Lemma 4: $2^{2k_1} = 2^t \leq x - 1$; or $2^{k_1} = 2^t \cdot 2^{v_2(s+1)} \leq 2^t(s+1) = (x-1) + 2^t \leq 2(x-1)$; or $2^{k_1} = 2^t \leq x - 1$. The first step is $q + 1 = 2^{k_1}x$, so

$$q + 1 \leq 2x(x-1) = 2x^2 - 2x.$$

Hence $q < 2x^2$, that is $x > \sqrt{q/2}$. □

Lemma 6 (consecutive triples). Let $a \to b \to c$ be consecutive elements of any orbit of C, with $qa + 1 = 2^J b$ and $qb + 1 = 2^K c$. Then

$$2^K ac - 2^J b^2 = a - b,$$

so if $a \neq b$ then $|a - b| \geq 2^{\min(J,K)}$. Consequently, if $J \leq K$ then $b > q$ when $b < a$, and $b^2 > qa$ when $b > a$.

Proof. Substitute $q = (2^K c - 1)/b$ into $qa + 1 = 2^J b$ and multiply by b. If $J \leq K$, then $|a - b| \geq 2^J = (qa+1)/b$; if $b > a$ this gives $b^2 > b(b-a) \geq qa + 1$, and if $b < a$ it gives $b(a-b) \geq qa + 1$, so $b > q$. □

Corollary 7. $y > x$ and $y^2 > qx$. In particular $2^{k_2} < q + 1$, so $2^{k_1+k_2} \leq 2^{k_1}q$.

Proof. Apply Lemma 6 to $x \to y \to 1$, so $J = k_2$ and $K = k_3$. If $k_3 < k_2$, then $|x - y| \geq 2^{k_3} = qy + 1 > y$, which forces $x > qy > q$, impossible as $x \leq (q+1)/2$. So $k_2 \leq k_3$. If $y < x$, Lemma 6 gives $y > q > x$, a contradiction. Hence $y > x$ and $y^2 > y(y-x) \geq qx + 1$. Finally $2^{k_2} = (qx+1)/y < (qx+1)/x < q + 1$, and 2^{k_2} is even while q is odd. □

4 The window and the near miss for x

Lemma 8 (the window). Let $N = 27 \cdot 2^h - (3q+1)^3$. Then

$$N = 9q(3 \cdot 2^{k_1} - 1) + 27 \cdot 2^{k_1+k_2} - 1, \qquad 0 < N < 54 \cdot 2^{k_1} q, \qquad 0 < \delta - \frac{1}{3} \le \frac{N}{81q^2} < \frac{2^{k_1+1}}{3q}.$$

Proof. The formula for N follows from Lemma 2 by expanding $(3q+1)^3$. Since $2^{k_1} \ge 2$, $N \ge 45q-1 > 0$, and $2^{k_1+k_2} \le 2^{k_1} q$ (Corollary 7) gives $N < 54 \cdot 2^{k_1} q$. Finally $N = 27\big[(q+\delta)^3 - (q+1/3)^3\big]$; as $N > 0$ we have $\delta > 1/3$, and then $N \ge 81q^2(\delta - 1/3)$. □

Lemma 9 (near miss for x). Let $f = h - 3k_1$. Then $3 \nmid f$, and

$$\left|x^3 - 2^f\right| \le \frac{3.2\, x^2}{2^{k_1}}, \qquad \left|x - 2^{f/3}\right| \le \frac{1.6}{2^{k_1}}.$$

Proof. Since $2^{3k_1} x^3 = (q+1)^3$, Lemma 2 gives $2^{3k_1} x^3 - 2^h = E$ with $E = 2q^2 + (3 - 2^{k_1})q + 1 - 2^{k_1+k_2}$. Using $q < 2^{k_1} x$, $|3 - 2^{k_1}| \le 2^{k_1}$ and $2^{k_1+k_2} \le 2^{k_1} q < 2^{2k_1} x$, we get $|E| \le 2 \cdot 2^{2k_1} x^2 + 2 \cdot 2^{2k_1} x + 1 \le 3.2 \cdot 2^{2k_1} x^2$ for $x \ge 3$. Dividing by 2^{3k_1} gives the first bound, and $f \equiv h \pmod 3$. Write $t = 2^{f/3}$. If $|x - t| \ge 1$, then $|x^3 - t^3| \ge 3x^2 - 3x + 1 \ge 2x^2 > 3.2x^2/2^{k_1}$, a contradiction since $2^{k_1} \ge 2$. So $|x - t| < 1$, hence $t > x - 1 \ge 2x/3$ and $x^2 + xt + t^2 > 2x^2$, which gives the second bound. □

5 Proof of Theorem 1

We use the following effective irrationality measure, which is the inequality obtained in the proof of Theorem 3 of Bauer and Bennett (2018) from their inequality (21), Lemma 7 and Proposition 8, before they round it to 1.454^{-n}.

Theorem A (Bauer–Bennett). Let $j \ge 11000$, $r \in \{1, 2\}$, $s \in \{1, 3\}$ and $n = 3j + r$. For every integer p,

$$\left|2^{r/3} - \frac{p}{s\, 2^j}\right| \ge \left|1 - \frac{p}{s\, 2^{n/3}}\right| > 2^{-5} \cdot 16166467234^{-(n+77)/63}.$$

The factor 2^{-5} is $1/(1.26 \cdot 2^{14/3})$ from their inequality (21); the paper's displayed intermediate step omits it, though it is absorbed in their final statement. Per unit of j the exponent is $\mu = 3\log_2(16166467234)/63 \approx 1.61487$.

Proof of Theorem 1. Suppose $q \ge 2^{22003}$. Write $a = k_1$, $h = 3m + r$ and $f = h - 3a = 3j' + r'$, where $r, r' \in \{1, 2\}$ by Lemmas 3 and 9.

1. **The window.** Apply Theorem A with $s = 3$, $p = 3q + 1$, $j = m$. Here $1 - p/(3 \cdot 2^{h/3}) = (\delta - 1/3)/2^{h/3}$, so $\delta - 1/3 > 2^m \cdot 2^{-5} \cdot 16166467234^{-(h+77)/63}$. With Lemma 8 and $q \ge 2^m$,

$$2^a > 1.5 \cdot 2^{2m-5} \cdot 16166467234^{-(h+77)/63}.$$

2. **The near miss for x.** Apply Theorem A with $s = 1$, $p = x$, $j = j'$. By Lemma 9, $|1 - x/2^{f/3}| \le 1.6/(2^a \cdot 2^{f/3})$, so

$$2^a < 1.6 \cdot 2^{5-j'} \cdot 16166467234^{(f+77)/63}.$$

3. **Combining.** By Lemma 9, $x \geq 2^{f/3} - 1.6/2^a \geq 2^{j'}$, so $q+1 = 2^a x \geq 2^a \cdot 2^{j'}$ and $m \geq a + j' - 1$. With $h = f + 3a$, step 1 gives $a > 0.20879\, f - 77.3$ and step 2 gives $a < 0.20496\, f + 47.2$. These are incompatible once $f \geq 32{,}214$, that is $j' \geq 10{,}738$.

Both applications of Theorem A need $j', m \geq 11000$, which holds because $x > \sqrt{q/2} \geq 2^{11001}$ by Lemma 5. So any 3-odd cycle through 1 has $q < 2^{22003}$. By Lemma 3 each h gives a single candidate $q = \lfloor 2^{h/3} \rfloor$, and checking every $h < 66{,}100$ directly (Appendix D) finds no cycle. □

Why the golden ratio appears. The proof squeezes x between two bounds, both governed by the exponent μ in Theorem A.

- **The window keeps x small.** By Lemma 8, $2^{h/3}$ exceeds $q + 1/3$ by about $1/(3x)$, so $(3q+1)/(3 \cdot 2^m)$ approximates $2^{r/3}$ with error about $1/(xq)$. Theorem A forbids errors below about $q^{-\mu}$, so $x \lesssim q^{\mu-1}$.
- **The near miss for x keeps x large.** By Lemma 9, $x/2^{j'}$ approximates $2^{r'/3}$ with error about $1/q$, where $2^{j'} \asymp x$. Theorem A forbids errors below about $x^{-\mu}$, so $x \gtrsim q^{1/\mu}$.

Both bounds can hold only if $1/\mu \leq \mu - 1$, that is $\mu^2 - \mu - 1 \geq 0$, whose positive root is $\varphi = (1+\sqrt{5})/2$. With $\mu = 1.61487$ the bounds read $q^{0.6192} \lesssim x \lesssim q^{0.6149}$, which are incompatible once q is large; the explicit constants in steps 1 to 3 make this precise for $q \geq 2^{22003}$. The two bounds are tied together by $q + 1 = 2^{k_1} x$: a large x makes the window's approximation too good, while a small x forces 2^{k_1} to be large and makes x's own approximation too good.

The threshold φ comes from this argument; the exponent comes from Bauer and Bennett, who needed only an exponent below 2 for their own purposes. Their stated exponent 1.62 lies just above φ and would leave room for x; the unrounded 1.61487 lies just below it.

6 Longer cycles

Lemma 10 (the ℓ-th power window). For an ℓ-odd cycle through 1,

$$2^h = (q+1) \prod_{i=1}^{\ell-1} \left(q + \frac{1}{x_i} \right).$$

If $\ell \geq 2$, then $q^\ell < 2^h < (q+1)^\ell$, $\ell \nmid h$ and $q = \lfloor 2^{h/\ell} \rfloor$. Moreover, with $\sigma = \sum 1/x_i$, we have $\delta - 1/\ell = \sigma/\ell + O(1/q)$.

Proof. Multiply the relations $2^{k_i} = (q \cdot x_{i-1} + 1)/x_i$ around the cycle. The product on the right is empty when $\ell = 1$, and then $2^h = q + 1$. If $\ell \geq 2$, the product contains a factor $q + 1/x_i$ with $x_i \geq 3$, and every such factor lies strictly between q and $q+1$. Hence $q^\ell < 2^h < (q+1)^\ell$. A power of 2 strictly between consecutive ℓ-th powers is not an ℓ-th power, so $\ell \nmid h$, and $q = \lfloor 2^{h/\ell} \rfloor$. Comparing $(q+\delta)^\ell$ with the expanded product gives the last statement. □

Lemma 11 (every element is a near miss). For $1 \leq i \leq \ell - 1$,

$$x_i \, 2^{S_i} = q^i (1 + \varepsilon_i), \qquad \varepsilon_i = \frac{1}{q} + \sum_{u=1}^{i-1} \frac{1 + \varepsilon_u}{q x_u}.$$

Consequently $x_i = 2^{n_i/\ell}(1+\rho_i)$ with $n_i = ih - \ell S_i$, and for $q \geq \ell$,

$$-\frac{i}{q+1} < \rho_i \leq \frac{2i+1}{3q}, \qquad \text{so} \qquad |\rho_i| \leq \frac{\ell-1}{q}.$$

If ℓ divides n_i, then $x_i \geq q/(\ell-1) - 1$.

Proof. The formula follows by induction from $q \cdot x_{i-1} + 1 = 2^{k_i} \cdot x_i$, using $2^{S_u}/q^{u+1} = (1+\varepsilon_u)/(q \cdot x_u)$. Then $1 + \rho_i = x_i \cdot 2^{S_i}/2^{ih/\ell} = (q/(q+\delta))^i (1+\varepsilon_i)$ with $0 < \delta < 1$. Since $x_1, \ldots, x_{i-1}$ are distinct odd numbers at least 3, their reciprocals sum to at most $(i-1)/3$, so by induction $\varepsilon_i \leq 1/q + (2/q)(i-1)/3 = (2i+1)/(3q)$, which is at most 1 for $q \geq \ell$. Also $1 \geq (q/(q+\delta))^i > (1 - 1/(q+1))^i \geq 1 - i/(q+1)$. Together these give the bounds on ρ_i, and both sides are at most $(\ell-1)/q$ in absolute value because $i \leq \ell - 1$. If $\ell \mid n_i$, the target $2^{n_i/\ell}$ is a power of 2 and x_i is odd, so $|x_i - 2^{n_i/\ell}| \geq 1$ and hence $|\rho_i| \geq 2^{-n_i/\ell} \geq (1 - |\rho_i|)/x_i$, which gives $x_i \geq 1/|\rho_i| - 1 \geq q/(\ell-1) - 1$. □

Theorem 2 is the case $R = \{1, \ldots, \ell - 1\}$ of the following.

Theorem 2′ (precise form of Theorem 2). Let $\ell \geq 2$, let $R \subseteq \{1, \ldots, \ell - 1\}$, and let $\varphi = (1 + \sqrt{5})/2$. Suppose that there are real numbers $0 < \mu < \varphi$, $c > 0$ and j_0 such that

$$\left| 2^{r/\ell} - \frac{p}{s \cdot 2^j} \right| > c \cdot 2^{-\mu j}$$

for all $r \in R$, $s \in \{1, \ell\}$, $p \in \mathbb{Z}$ and integers $j \geq j_0$. Then there is a number Q, computable from ℓ, μ, c and j_0, with the following property. If $q > Q$ is odd and $1 = x_0 \to x_1 \to \cdots \to x_{\ell-1} \to 1$ is a cycle of C of length ℓ, with $q \cdot x_{i-1} + 1 = 2^{k_i} \cdot x_i$ for $1 \leq i \leq \ell$ (where $x_\ell = 1$), then some multiple of $h = k_1 + \cdots + k_\ell$ has a nonzero residue modulo ℓ outside R.

For $R = \{1, \ldots, \ell - 1\}$ the conclusion is that no $q > Q$ has 1 on a cycle of length ℓ. For $\ell = 4$ and $R = \{2\}$, the multiples of h avoid the residues 1 and 3 exactly when h is even, so the conclusion is that h is odd.

Proof of Theorem 2′. Suppose that every multiple of h has its nonzero residues modulo ℓ in R; we show that q is bounded. The window uses $2^{r/\ell}$ with $r \equiv h$, and the element x_i uses $r \equiv n_i \equiv ih \pmod{\ell}$, so every root that occurs below has $r \in R$. By Lemma 11 and the hypothesis, each x_i with $\ell \nmid n_i$ satisfies $c \cdot 2^{-\mu j} < |2^{r/\ell} - x_i/2^j| \ll 1/q$ with $2^j \asymp x_i$, so $x_i \gg q^{1/\mu}$; the others satisfy $x_i \gg q$. (Elements with j below the hypothesis's threshold are bounded, and each has a fixed nonzero relative error, so by Lemma 11 they occur only for bounded q.) Hence $\sigma \ll q^{-1/\mu}$ and, by Lemma 10, $|\delta - 1/\ell| \ll q^{-1/\mu}$. On the other hand

$$\delta - \frac{1}{\ell} = 2^m \left(2^{r/\ell} - \frac{\ell q + 1}{\ell \cdot 2^m} \right)$$

with $h = \ell m + r$, so the hypothesis gives $|\delta - 1/\ell| \gg q^{1-\mu}$. Thus $q^{1-\mu} \ll q^{-1/\mu}$, which bounds q exactly when $\mu^2 - \mu - 1 < 0$. □

For $\ell = 3$, Theorem A supplies the hypothesis of Theorem 2′ with $R = \{1, 2\}$, $\mu \approx 1.61487$ and $j_0 = 11000$. So Theorem 1 is the case $\ell = 3$ of Theorem 2; Section 5 carries out this case with explicit constants.

Ridout's theorem (1958). Let α be an algebraic irrational, let S be a finite set of primes, and let $\varepsilon > 0$. Only finitely many reduced fractions p/q with $q > 0$ have every prime factor of q in S and satisfy

$$\left| \alpha - \frac{p}{q} \right| < q^{-1-\varepsilon}.$$

The theorem gives no bound on those fractions. This is why Theorem 3, which uses it in place of an effective measure, gives no bound on q.

Theorem 3. For each $\ell \geq 2$, only finitely many odd q have 1 on a cycle of C of length ℓ.

Proof of Theorem 3. Let $r \in \{1, \ldots, \ell - 1\}$ and let S consist of 2 and the prime factors of ℓ. Then $2^{r/\ell}$ is an algebraic irrational. A fraction $p/(s \cdot 2^j)$ with $s \in \{1, \ell\}$, written in lowest terms, has every prime factor of its denominator in S. Ridout's theorem therefore gives, for any $\varepsilon > 0$,

$$\left|2^{r/\ell} - \frac{p}{s \cdot 2^j}\right| > (s \cdot 2^j)^{-1-\varepsilon}$$

for all but finitely many such fractions. The proof of Theorem 2′ runs with this inequality in place of an effective measure, and with $\mu = 1 + \varepsilon < \varphi$. Each exceptional fraction has a fixed nonzero distance from $2^{r/\ell}$. For an element x_i this fixes x_i and its relative error, which Lemma 11 allows only for bounded q, and for the window it is incompatible with $|\delta - 1/\ell| \to 0$. □

Theorem 3 fails for $\ell = 1$. Evidently, $q = 2^e - 1$ gives the cycle $1 \to 1$ for every $e \geq 2$. Mathematically, for $\ell \geq 2$ an odd element other than 1 puts a factor strictly between q and $q + 1$, so ℓ does not divide h and $2^{r/\ell}$ is irrational. Ridout's theorem then applies. For $\ell = 1$ that product is empty, so the same identity becomes $2^h = q + 1$, and the three conclusions of Lemma 10 fail. The root is the integer 2^h, and Ridout's theorem does not apply.

Corollary 12. No odd q has 1 on a cycle of C of length 4 whose total number of halvings is $\equiv 2 \pmod 4$.

Proof. This is Theorem 2′ with $\ell = 4$ and $R = \{2\}$, with explicit constants. Write $h = 4m + 2$. Then $n_1 \equiv n_3 \equiv 2$ and $n_2 \equiv 0 \pmod 4$, so only $\sqrt{2}$ and powers of 2 occur, and the window approximates $\sqrt{2}$ by $(4q + 1)/2^{m+2}$. Bauer and Bennett's Corollary 1.6 (2002) gives $|\sqrt{2} - p/2^j| > 2^{-1.48j}$ for every $j > 2$ with $j \notin \{3, 7, 8\}$.

- By Lemma 11, $x_2 \geq q/3.1$, and x_1, x_3 are near $\sqrt{2} \cdot 2^j$, so $4.3/q > (x/1.41)^{-1.48}$, giving $x_1, x_3 \geq (0.387q)^{0.6757}$.
- The measure gives $|\delta - 1/4| > 0.152(q + 1)^{-0.48}$, while expanding the product in Lemma 10 explicitly gives $|\delta - 1/4| \leq \sigma/4 + 1.75/q \leq 0.95\, q^{-0.6757} + 2.53/q$.

For $q \geq 10^5$ every exponent involved exceeds 8, so Corollary 1.6 applies, and the two bounds are then incompatible (the inequalities already conflict from $q = 2 \times 10^4$). The search over h covers $\ell = 4$ for $q < 2^{25{,}000}$. □

Corollary 13. Let $\ell = 2^a$ with $a \geq 2$. Only finitely many odd q have 1 on a cycle of C of length ℓ whose total number of halvings is $\equiv \ell/2 \pmod \ell$, and a bound is computable from Bauer and Bennett's Corollary 1.6 (2002). For $\ell = 4$ every such q is ruled out by Corollary 12.

Proof. Write $h = \ell m + \ell/2$. Every multiple of h is congruent to 0 or $\ell/2$ modulo ℓ, so the only root that occurs is $2^{1/2}$. A denominator $s \cdot 2^j$ with $s \in \{1, \ell\}$ is a power of 2, because ℓ is. Bauer and Bennett's Corollary 1.6 (2002) therefore gives the hypothesis of Theorem 2′ with $R = \{\ell/2\}$, $\mu = 1.48 < \varphi$, and $j_0 = 9$: the finitely many excluded exponents $j \in \{1, 2, 3, 7, 8\}$ contribute only a constant factor depending on ℓ. Theorem 2′ supplies the computable bound. The case $\ell = 4$ is Corollary 12. □

Corollary 14. Let $\ell = 3 \cdot 2^a$ with $a \geq 1$. Only finitely many odd q have 1 on a cycle of C of length ℓ whose total number of halvings is $\equiv \ell/3$ or $2\ell/3 \pmod{\ell}$, and a bound is computable from Theorem A. The length 3 is Theorem 1, which rules out every such q.

Proof. Write $h = \ell m + r_0$ with $r_0 \in \{\ell/3,\, 2\ell/3\}$. Every multiple of h is congruent to 0, $\ell/3$, or $2\ell/3$ modulo ℓ, so the only irrational roots that occur are $2^{1/3}$ and $2^{2/3}$. In the proof of Theorem 2′ the window fraction is $(\ell q + 1)/(\ell \cdot 2^m)$. Its numerator is coprime to ℓ, and $\ell \cdot 2^m = 3 \cdot 2^{m+a}$, so this is a fraction of the shape in Theorem A with $s = 3$ and exponent $m + a$. Each near miss is $x_i/2^j$ with x_i odd, so Theorem A applies with $s = 1$. Both give the exponent $\mu = 3\log_2(16166467234)/63 \approx 1.61487 < \varphi$ once the exponent in Theorem A is at least 11000. The shift by a, and the fixed factors in Theorem A, contribute only a constant depending on ℓ. The fractions used in the proof of Theorem 2′ therefore satisfy its hypothesis with $R = \{\ell/3,\, 2\ell/3\}$, this μ, and $j_0 = 11000$, and that proof supplies the computable bound. □

7 Other results

The following were found along the way. They are not needed for Theorems 1 to 3 and Corollaries 12 and 13; full statements and proofs are in the appendices indicated, and Appendix E has notes on the literature and the history of the work.

- **One and two odd steps** (Appendix A). The two-step case, only $q = 5$, follows from Beukers's theorem on $x^2 + D = 2^n$.
- **The order of** 2 (Appendix B.1). $k_3 = d$ or $2d$, where d is the order of 2 modulo q; in particular $d < 3\log_2 q$, and no Wieferich prime q has a 3-odd cycle through 1, the same mechanism Franco and Pomerance use for Crandall's question of when orbits avoid 1.
- **Finiteness for three steps via Ridout alone** (Appendix B.2). Lemmas 5 and 8 bound the window by about $q^{-1/2}$, and Ridout's theorem then gives finiteness. This is the case $\ell = 3$ of Theorem 3.
- **Elementary special cases** (Appendix B.3). $k_1 = k_2 = 1$ is impossible via the integral points of $Y^2 = X^3 - 4X^2 + 12X + 16$ ($q = 41$ is a near miss), and $k_1 = 1$, $k_2 = 2$ via the factorization $(q + 2)(q^2 - q + 4)$.
- **Bounds on** k_1 (Appendix B.4). Bauer and Bennett's inequalities give $q^{1/3}/2 < 2^{k_1} < 2.01(q + 1)^{2/5}$, so each k_1 allows only q in an explicit finite range, and $k_1 = 1$ is impossible.
- **The cases of Lemma 4** (Appendix B.5). The cases $t > k_1$ and $t = k_1$ are impossible, and the case $t < k_1$ is tightly constrained.
- **Small elements and an earlier bound on** y (Appendix B.6). In fact $x \geq 5$ and $y \geq 7$. Before Lemma 6, the best result was $y \geq x$ or $y > (3q)^{1/3}$; this is the origin of the threshold $q \geq 71$ in earlier versions of the window lemma.
- **An earlier approach to four odd steps** (Appendix C), reducing finiteness to lower bounds on the elements, with $x > \sqrt{q/2}$ proved and two configurations left open.
- **Computations** (Appendix D). Searches over h for cycles through 1 with $\ell = 3$ to 15, a search for an arbitrary cycle with $\ell = 2$ to 9 and for odd q from 5 through 25 at higher lengths, verification of the Bauer–Bennett inputs, and the digit scan ruling out $q < 2^{1{,}071{,}999{,}953}$.

8 Open problems

1. **Four steps with h odd.** The window and the near misses for x_1 and x_3 involve $2^{1/4}$ and $2^{3/4}$. Theorem 2 would finish the case given an effective exponent below φ for these numbers with power-of-2 denominators. The Bauer–Bennett construction needs a fourth power exceptionally close to a power of 2; among odd $a \le 3 \times 10^6$ the best is $19^4 = 2^{17} - 751$, of quality $\log_2(2^{17})/\log_2 751 \approx 1.78$, against 4.4 for the cube case ($5^3 + 3 = 2^7$) and 5.3 for $\sqrt{2}$ ($181^2 + 7 = 2^{15}$). A different construction is likely needed.

2. **Effective results for larger ℓ.** Theorem 2 turns any such exponent for the roots $2^{r/\ell}$ into a complete answer for that ℓ. Santos's conjecture in full would need this for every ℓ, or a bound uniform in ℓ.

3. **Novelty.** Franco's 1990 thesis on Diophantine approximation and the $qx + 1$ problem could not be consulted and should be checked for overlap.

A One and two odd steps

One odd step. The cycle $1 \to 1$ requires $q + 1 = 2^e$, so $q = 2^e - 1$ (Santos, Proposition 15).

Proposition A.1 (two odd steps). The only odd q with 1 on a cycle of C of length 2 is $q = 5$.

Proof. The cycle gives $q(q+1) = 2^{k_1}(2^{k_2} - 1)$. Multiplying by 4 and adding 1,

$$(2q+1)^2 + (2^{k_1+2} - 1) = 2^{k_1+k_2+2}.$$

This is $X^2 + D = 2^n$ with $X = 2q+1$ and $D = 2^{k_1+2} - 1$. By a theorem of Beukers (Theorem 1.8 in Bauer and Bennett 2002), for $D \neq 7$, 23 and D not of the form $2^k - 1$ there is at most one solution, while for $D = 2^k - 1$ with $k \ge 4$ there are exactly two, $X = 1$ and $X = 2^{k-1} - 1$. These give $q = 0$ and $q = 2^{k_1} - 1$, the one-step cycle traversed twice, so $k_1 \ge 2$ gives nothing new. For $k_1 = 1$, $D = 7$ is the Ramanujan–Nagell equation with $X \in \{1, 3, 5, 11, 181\}$, and only $X = 11$ gives an odd q other than 1: $q = 5$, with $1 \to 3 \to 1$. □

As a check, the argument of Lemma B.2 below gives $2^{k_2} = qx + 1 < q^2 < 2^{2d}$ with d the order of 2 modulo q, so $k_2 = d$, and the cycle condition reads $q(q+1) = 2^{k_1}(2^d - 1)$; for $q = 5$, $d = 4$ and $5 \cdot 6 = 2 \cdot 15$.

B Earlier results on three odd steps

These results were found before Theorem 1 and are not needed for it. Notation is as in Sections 3 to 5. Bauer and Bennett's headline inequality, $|x^3 - 2^n| \ge x^{4/3}$ unless $x^3 = 2^n$ or $(x, n) = (5, 7)$, and their Corollary 4, $|x^3 - 27 \cdot 2^n| \ge 3^{5/3} x^{4/3}$ unless $x^3 = 27 \cdot 2^n$ or $x \in \{4, 5, 8, 15, 19, 38, 121\}$, are used throughout; both were also checked numerically for $n < 200$ and $n < 70$ respectively.

B.1 Factorization and the order of 2

Lemma B.1 (factorization). With $q = 2^{k_1}x - 1$,

$$(2^{k_1}x - 1)(2^{k_1}x^2 - x + 1) = 2^{k_2}(2^{k_3} - 1).$$

Proof. The factors are q and $qx + 1$, and $q(qx + 1) = q \cdot 2^{k_2}y = 2^{k_2}(2^{k_3} - 1)$. □

Completing the square in $qx + 1 = 2^{k_2}y$ gives $(2q + 1)^2 + (2^{k_1+2} - 1) = 2^{k_1+k_2+2} \cdot y$, so the three-step case is the equation of Proposition A.1 with an odd cofactor y, tied to q by $qy + 1 = 2^{k_3}$.

Lemma B.2 (order of 2). $k_3 = d$ or $k_3 = 2d$, where d is the order of 2 modulo q. In particular $d < 3 \log_2 q$.

Proof. $2^{k_3} = qy + 1 \equiv 1 \pmod q$, so d divides k_3. Using $2^{k_1} \geq 2$ and $2^{k_2} \geq 2$, $y \leq (q^2 + q + 2)/4$, so $2^{k_3} \leq q(q^2 + q + 2)/4 + 1 < q^3$. Also $2^d \geq q + 1 > q$, so $q^3 < 2^{3d}$, and hence $k_3 < 3d$. □

Corollary B.3 (Wieferich primes). Let $F = (2^d - 1)/q$ and $k_3 = md$. Then $mF \equiv 2^{-k_2} \pmod q$, so $\gcd(F, q) = 1$. In particular no Wieferich prime q ($q^2 \mid 2^{q-1} - 1$) has a 3-odd cycle through 1.

Proof. Since $2^d \equiv 1 \pmod q$, $y = (2^{md} - 1)/q = F(1 + 2^d + \cdots + 2^{(m-1)d}) \equiv mF \pmod q$. Reducing $2^{k_2}y = qx + 1 \bmod q$ gives $2^{k_2} \cdot mF \equiv 1$. For a Wieferich prime, the lifting-the-exponent lemma shows that q^2 divides $2^d - 1$, since q^2 divides $2^{q-1} - 1$ and q does not divide $(q - 1)/d$; so $q \mid F$. □

This is the mechanism Franco and Pomerance use for Crandall's question of when orbits avoid 1. Together with Lemma 3, Lemma B.2 determines everything from q: $k_1 = v_2(q + 1)$, h is the unique exponent with $q^3 < 2^h < (q + 1)^3$, $k_3 \in \{d, 2d\}$ and $k_2 = h - k_1 - k_3$.

B.2 Finiteness via Ridout's theorem

Proposition B.4. Only finitely many odd q have a 3-odd cycle through 1.

Proof. By Lemmas 8 and 5, $\delta - 1/3 < 2^{k_1+1}/(3q)$ with $2^{k_1} = (q + 1)/x < (q + 1)\sqrt{2/q}$, so $\delta - 1/3 < 0.96\, q^{-1/2}$ for $q \geq 71$. With $h = 3m + r$, $\beta = 2^{r/3}$, $P = 3q + 1$ and $Q = 3 \cdot 2^m$ this gives $0 < \beta - P/Q < 2.9\, q^{-1/2}/Q$, and since $Q < 2.39(q + 1)$ we get $|\beta - P/Q| < 4.6\, Q^{-3/2} < Q^{-5/4}$ for $Q > 450$. The reduced fraction still has a denominator built from 2 and 3, so Ridout's theorem with $S = \{2, 3\}$ and $\varepsilon = 1/4$ allows only finitely many such fractions; infinitely many q would give infinitely many distinct ones, since the errors tend to 0 and β is irrational. □

This is the case $\ell = 3$ of Theorem 3, and Theorem 1 supersedes it. The exponent $3/2$ comes from the lower bounds $x > \sqrt{q/2}$ and $y > x$; with only $x, y \geq 3$ the error is of order $1/Q$ and Ridout's theorem gives nothing. The result is ineffective, which is why Theorem 1 needs Theorem A instead.

B.3 Elementary special cases

Proposition B.5 ($k_1 = k_2 = 1$). No 3-odd cycle through 1 has $k_1 = k_2 = 1$. Hence $q \not\equiv 1 \pmod 8$.

Proof. Lemma 2 gives $2^h = q^3 + q^2 + 2q + 4 = (q+1)(q^2+2) + 2$. With $q + 1 = 2x$ this becomes

$$4x^3 - 4x^2 + 3x + 1 = 2^n, \qquad n = h - 1,$$

and $q^2 + 2 = 4x^2 - 4x + 3$ divides $2^n - 1$. Since q is odd, $q^2 + 2 \equiv 3 \pmod 8$. Every prime p dividing $q^2 + 2$ has -2 as a square mod p, so $p \equiv 1$ or $3 \pmod 8$, and at least one is $\equiv 3 \pmod 8$. For that p, 2 is not a square mod p, so the order of 2 modulo p is even and divides n. So n is even, and $(X, Y) = (4x,\ 4 \cdot 2^{n/2})$ is an integral point on

$$E\colon Y^2 = X^3 - 4X^2 + 12X + 16.$$

Using Sage (passagemath) with proofs enabled, E has rank 2 and trivial torsion, generated by $(0, 4)$ and $(1, 5)$ with saturation index 1, and its integral points are $X \in \{0, 1, 4, 9, 84\}$, computed by the elliptic-logarithm method. $X = 4$ gives $q = 1$; $X = 84$ gives $x = 21$ but $Y = 752 = 4 \cdot 188$, and 188 is not a power of 2; the others do not have the form $4x$ with x odd and at least 3. For the congruence: if $q \equiv 1 \pmod 8$, then $k_1 = 1$ and $x \equiv 1 \pmod 4$, so $t > k_1$ in Lemma 4, and its proof forces $k_2 = k_1 = 1$. □

The point $X = 84$ is a near miss: for $q = 41$, $q^3 + q^2 + 2q + 4$ is twice a perfect square, and the cycle fails only at the last step, since $41 \cdot 431 + 1 = 17{,}672 = 8 \cdot 47^2$.

Proposition B.6 ($k_1 = 1$, $k_2 = 2$). No 3-odd cycle through 1 has $k_1 = 1$ and $k_2 = 2$. Hence $q \not\equiv 13 \pmod{16}$.

Proof. Lemma 2 gives $2^h = q^3 + q^2 + 2q + 8 = (q+2)(q^2 - q + 4)$, and $q + 2$ is an odd number at least 5. For $k_1 = 1$, $k_2 = 2$ holds exactly when $x \equiv 7 \pmod 8$, that is $q \equiv 13 \pmod{16}$. □

Both are superseded by Corollary B.8, which rules out $k_1 = 1$ altogether.

B.4 Effective bounds on k_1

Proposition B.7. In any 3-odd cycle through 1, $2^{k_1} > q^{1/3}/2$.

Proof. Apply Bauer and Bennett's Corollary 4 with $x = 3q + 1$ and $n = h$. The exceptional values of x give $q \in \{1, 6, 40\}$, which are excluded, and $N > 0$ by Lemma 8, so $N \geq 3^{5/3}(3q+1)^{4/3} > 27\, q^{4/3}$. By Lemma 8, $N < 54 \cdot 2^{k_1} q$. Comparing, $54 \cdot 2^{k_1} q > 27\, q^{4/3}$. □

Corollary B.8. (a) $k_1 = 1$ is impossible: it would force $q < 64$, and the direct search excludes every $q < 2{,}000{,}000$; so every possible q is $\equiv 3 \pmod 4$. (b) Together with Lemma 5, $2^{2k_1 - 1} - 1 < q < 2^{3k_1 + 3}$, so each k_1 allows only an explicit finite range of q. (c) Since the digit scan (Appendix D) excludes $q < 2^{1{,}071{,}999{,}953}$, any remaining q would need $k_1 \geq 357{,}333{,}317$, so $q \equiv -1 \pmod{2^{357{,}333{,}317}}$.

Theorem 1 rules out these q altogether.

Proposition B.9 (upper bound on k_1). $2^{k_1} \leq 3.2\, x^{2/3}$, and hence $2^{k_1} < 2.01(q+1)^{2/5}$.

Proof. By Lemma 9, $x^3 - 2^f = E/2^{3k_1} \neq 0$ with $|E| \leq 3.2 \cdot 2^{2k_1}x^2$, so Bauer and Bennett's headline inequality gives $2^{3k_1}x^{4/3} \leq 3.2 \cdot 2^{2k_1}x^2$. The exception $x = 5$ forces $q \leq 39$ by Lemma 5. □

Together with Proposition B.7, $q^{1/3}/2 < 2^{k_1} < 2.01(q+1)^{2/5}$.

B.5 The three cases of Lemma 4

Proposition B.10 ($t > k_1$ is impossible). No 3-odd cycle through 1 has $t > k_1$.

Proof. Lemma 4 and its proof give $t = 2k_1$ and $k_2 = k_1$, so $x = 1+2^{2k_1}s$ with s odd. Proposition B.7 gives $q < 8 \cdot 2^{3k_1}$, so $x < 8 \cdot 2^{2k_1}$ and $s \in \{1, 3, 5, 7\}$. Then $q = s \cdot 2^{3k_1} + 2^{k_1} - 1$, and the cycle equation reads $2^h = q^3 + q^2 + 2^{k_1}q + 2^{2k_1}$. Dividing by 2^{9k_1},

$$s^3 < \frac{2^h}{2^{9k_1}} < \left(s + \frac{1}{2^{2k_1}}\right)^3 \left(1 + \frac{2}{q}\right),$$

and for $2^{k_1} \geq 8$ the right-hand side is below the next power of 2 after s^3 (2, 32, 128, 512), so $2^h/2^{9k_1}$ lies strictly between consecutive powers of 2. For $2^{k_1} \leq 4$, $q \leq 451$, which the direct search covers. □

(Also checked in exact arithmetic for $k_1 < 400$.)

Proposition B.11 ($t = k_1$ is impossible). No 3-odd cycle through 1 has $t = k_1$.

Proof. Here $x = 1 + 2^{k_1}s$ with s odd, so $q = 2^{2k_1}s + 2^{k_1} - 1$. Let $g = h - 6k_1$. Expanding the cycle equation,

$$0 < 2^g - s^3 = \frac{R}{2^{6k_1}} \leq \frac{6s^2}{2^{k_1}}, \qquad R = 3{\cdot}2^{4k_1}s^2(2^{k_1}{-}1) + 3{\cdot}2^{2k_1}s(2^{k_1}{-}1)^2 + (2^{k_1}{-}1)^3 + q^2 + 2^{k_1}q + 2^{k_1+k_2},$$

using $q < 2{\cdot}2^{2k_1}s$ and $2^{k_2} < q$ ($2^{k_1} \geq 4$). Bauer and Bennett's headline inequality gives $2^g - s^3 \geq s^{4/3}$, so $2^{k_1} \leq 6\,s^{2/3}$, while Proposition B.7 gives $8 \cdot 2^{3k_1} > q > 2^{2k_1}s$, so $2^{k_1} > s/8$. Hence $s < 48^3$ and $2^{k_1} \leq 13{,}824$, so $q < 2 \cdot 2^{2k_1}s < 5 \times 10^{13}$, which the computations cover. The exception $s = 5$ gives $q < 4 \times 10^6$ through Proposition B.9. □

Proposition B.12 (a constraint when $t < k_1$). Suppose $t < k_1$, so that $k_2 = t$, and write $x = 2^{k_1}u - (2^{k_2} - 1)$ with u odd. Then $2^{k_1}/2^{k_2} \leq 3u^{2/3}$.

Proof. Put $\delta' = 2^{k_1+k_2} - 2^{k_1} + 1$, so $q = 2^{2k_1}u - \delta'$, and $g = h - 6k_1$. Expanding,

$$0 < u^3 - 2^g = \frac{3 \cdot 2^{4k_1}u^2\delta' - 3 \cdot 2^{2k_1}u\delta'^2 + \delta'^3 - q^2 - 2^{k_1}q - 2^{k_1+k_2}}{2^{6k_1}} \leq \frac{3u^2 \cdot 2^{k_2}}{2^{k_1}},$$

since the terms after the first are negative in total and $\delta' < 2^{k_1+k_2}$. Bauer and Bennett's headline inequality gives $u^3 - 2^g \geq u^{4/3}$. The exceptions are $u = 1$, where $x \leq 2^{k_1}$ and Proposition B.9 forces $2^{k_1} \leq 32$, and $u = 5$, where it forces $q < 4 \times 10^6$. □

The positivity and upper bounds in Propositions B.11 and B.12 were also checked in exact arithmetic on hundreds of thousands of random parameter choices. Propositions B.10 and B.11 leave only the case $t < k_1$; Theorem 1 makes the case analysis unnecessary.

B.6 Small elements and an earlier bound on y

Remark (small elements). The elements x and y are odd and different from 1, so $x, y \geq 3$, and Corollary 7 gives $y \geq 5$. If $x = 3$, Lemma 5 gives $2^{k_1} \leq 2(x-1) = 4$, so $q = 3 \cdot 2^{k_1} - 1 \in \{5, 11\}$, and neither has a 3-odd cycle through 1. Hence $x \geq 5$ and $y \geq 7$. None of the proofs in the main text need more than $x \geq 3$ and $y > x$.

Before Lemma 6 was found, the case $y < x$ could not be excluded, and the following weaker bound was used instead. Corollary 7 shows that $y < x$ never happens, so the second alternative below is vacuous.

Proposition B.13. In a 3-odd cycle through 1, either $y \geq x$ or

$$y > \left(\frac{3q^2}{q+1}\right)^{1/3} \approx (3q)^{1/3}.$$

Moreover, in the second case $v_2(y-1) = k_1$ exactly.

Proof. Multiplying $2^{k_2} y = qx + 1$ by $2^{k_1} y^2$ and using $qy = 2^{k_3} - 1$ and $q + 1 = 2^{k_1} x$ gives the exact identity

$$2^{2k_3} + (y-2)\, 2^{k_3} + C = 2^{k_1+k_2} y^3, \qquad C = 2^{k_1} y^2 - y + 1.$$

If 2^{k_3} divides C, then $C \geq 2^{k_3} = qy + 1$, which simplifies to $2^{k_1} y \geq q + 1$, that is $y \geq x$. Otherwise the first two terms are divisible by a higher power of 2 than C, so $v_2(C) = k_1 + k_2$ and $2^{k_1+k_2} \leq C < 2^{k_1} y^2$. Since the left side exceeds $2^{2k_3} > q^2 y^2$, this gives $q^2 y^2 < 2^{k_1+k_2} y^3 < 2^{k_1} y^5$, so $q^2 < 2^{k_1} y^3 \leq (q+1) y^3/3$. For the last claim, $y \equiv 1 \pmod{2^{k_1}}$ by Lemma 1; if $v_2(y-1) > k_1$, then $v_2(C) = k_1$, forcing $k_2 = 0$, which is impossible. □

For $q \geq 71$ the cube root is the smaller of the two bounds, so this gave $y > 1.436\, q^{1/3}$ unconditionally, and for $q > 2{,}000{,}000$ it gave $y \geq 183$. The identity was also checked by computer algebra: substituting the cycle relations into both sides and expanding gives the same expression.

C An earlier approach to four odd steps

Before Theorem 3, finiteness for ℓ odd steps was reduced to lower bounds on the elements of the cycle.

Proposition C.1 (reduction to lower bounds). Fix ℓ. If some $\theta > 0$ and $c > 0$ make every element other than 1 of every ℓ-odd cycle through 1 at least $c \cdot q^{\theta}$, then only finitely many q have such a cycle.

Proof. As for Proposition B.4, with $h = \ell m + r$, $\beta = 2^{r/\ell}$, $P = \ell q + 1$ and $Q = \ell \cdot 2^m$: Lemma 10 gives $|\beta - P/Q| \ll Q^{-1-\theta}$, and Ridout's theorem applies with $S = \{2\} \cup \{\text{primes dividing } \ell\}$. □

Theorem 3 makes this unnecessary, since Lemma 11 supplies the lower bounds directly. For four odd steps, written $1 \to x \to y \to z \to 1$, the following partial results were obtained.

Lemma C.2. In a 4-odd cycle through 1, $2^{k_1} \leq 2(x-1)$ and $x > \sqrt{q/2}$.

Proof. As in Lemma 1, $z \equiv 1 \pmod{2^{k_1}}$, so $qy + 1 = 2^{k_3} z$ gives $y \equiv 1 - 2^{k_3} \pmod{2^{k_1}}$. Let $t = v_2(x-1)$ and $s = (x-1)/2^t$. If $t \geq k_1$, then $2^{k_1} \leq x - 1$. Otherwise $k_2 = t$ and $y = 2^{k_1 - t} x^2 - s \equiv 2^{k_1 - t} - s \pmod{2^{k_1}}$, since $x^2 \equiv 1 \pmod{2^t}$. So $M = 2^{k_1 - t} + 2^{k_3} - (s+1)$ is a multiple of 2^{k_1}. If $M = 0$, then $2^{k_1 - t} < s + 1$, so $2^{k_1} < 2^t (s+1) \leq 2(x-1)$. If $M \leq -2^{k_1}$, then $s + 1 > 2^{k_1}$, so $2^{k_1} \leq x$. If $M \geq 2^{k_1}$, then $2^{k_3} > 2^{k_1 - 1}$, so $k_3 \geq k_1$ and $2^{k_1 - t} \equiv s + 1 \pmod{2^{k_1}}$, giving $2^{k_1 - t} = s + 1$ or $s + 1 \geq 2^{k_1}$; either way $2^{k_1} \leq 2(x-1)$. The first step is $q + 1 = 2^{k_1} x$, so

$$q + 1 \leq 2x(x-1) = 2x^2 - 2x.$$

Hence $q < 2x^2$, that is $x > \sqrt{q/2}$. □

Lemma 6 then gives $y \gg q^{3/4}$ if $k_2 \leq k_3$, and $z > q$ or $z^2 > qy$ if $k_3 \leq k_4$. Two configurations were left open: $k_3 < k_2$ with $y < x$, where y could be small, and $k_4 < k_3$, where z could be small. With Proposition C.1 these would have given finiteness for four steps; Theorem 3 gives it directly.

D Computations

Searches. By Lemma 10 each h gives one candidate $q = \lfloor 2^{h/\ell} \rfloor$, which is tested by following 1 for ℓ odd steps. For $\ell = 3$ the same test was run for every $h \leq 200{,}000$. It was repeated independently for every $h < 66{,}100$, which covers the range $q < 2^{22003}$ needed in Theorem 1. For each ℓ from 4 to 15, the same test was run for every $h \leq 100{,}000$ with $\ell \nmid h$.

odd steps ℓ	no cycle for q below
3	$2^{66{,}666}$
4	$2^{25{,}000}$
5	$2^{20{,}000}$
6	$2^{16{,}666}$
7	$2^{14{,}285}$
8	$2^{12{,}500}$
9	$2^{11{,}111}$
10	$2^{10{,}000}$
11	$2^{9{,}090}$
12	$2^{8{,}333}$
13	$2^{7{,}692}$
14	$2^{7{,}142}$
15	$2^{6{,}666}$

Known cycles. Few cycles of C are known. The cycle $1 \to 1$ occurs precisely when $q = 2^e - 1$. For $q = 5$ and $h = 5$ there is $1 \to 3 \to 1$. For $q = 5$ and $h = 7$ there are $13 \to 33 \to 83 \to 13$ and $17 \to 43 \to 27 \to 17$. For $q = 181$ and $h = 15$ there are $27 \to 611 \to 27$ and $35 \to 99 \to 35$.

A search for any cycle. The searches above follow the orbit of 1. The same constraint on q holds for a cycle of any odd elements. Let the elements of such a cycle be $x_0, \ldots, x_{\ell-1}$, with $\ell \geq 2$. The product of the steps is $2^h = \prod_{i=0}^{\ell-1} (q + 1/x_i)$. Each factor is greater than q, and some element is at least 3, so the corresponding factor is at most $q + 1/3$. Hence $q^\ell < 2^h < (q+1)^\ell$, so ℓ does not divide h and $q = \lfloor 2^{h/\ell} \rfloor$. For each ℓ from 2 to 9, every such cycle with h at most the bound below was tested. The only cycles in these ranges are the ones listed above.

odd steps ℓ	h at most	no other cycle for q below
2	8,000	$2^{4,000}$
3	4,000	$2^{1,333}$
4	3,000	2^{750}
5	2,000	2^{400}
6	1,800	2^{300}
7	700	2^{100}
8	560	2^{70}
9	270	2^{30}

With q fixed, the search runs through lengths and keeps only the values of h satisfying $q^\ell < 2^h < (q+1)^\ell$. If no integer h lies in that range, that length has no cycle. For $q = 5$, lengths 2 through 20 produce only $1 \to 3 \to 1$ and the two cycles of length 3. For each odd q from 7 through 25, lengths 2 through 16 produce no cycle.

Checks of Theorem A. $F(\tau) = 11.97804\ldots$ at $\tau = 0.20305\ldots$, and $9 \cdot 2^{28} \cdot F/1.79$ equals $16166467233.3\ldots$, matching the paper. The constant in (21) is $2^{-5.0001}$, as used above. Their Lemma 7 was checked at 3000-digit precision for every n_1 from 2 to 60, both δ and both r, and the side claim in its proof gives $\max f = 0.31999\ldots < 1/3$. Their Proposition 8 ($\Pi > 1.8^{n_1}$ for every $n_1 \geq 497$) was rechecked in full:

- **Lemma 9** (the primes that divide every Padé coefficient) was verified directly for every $n_1 \leq 160$, both δ and both r: 5,978 cases, no failures.
- **The lower bound** $L(r, n_1) \leq \log \Pi$ was computed exactly for every $497 \leq n_1 < 10^6$ and both r. It reproduces the paper's values $\exp(L(2, 3319)/3319) = 1.897735\ldots$ and $\exp(L(1, 10000)/10000) = 2.006523\ldots$. It exceeds $\log 1.8$ except for 43 pairs (n_1, r), all with $n_1 \leq 830$; for these Π was computed exactly for both δ, with minimum $\Pi^{1/n_1} = 1.80767$ at $n_1 = 497$. For $1000 < n_1 < 10^6$ the minimum of $\exp(L/n_1)$ is 1.8422.
- **The analytic estimates** for $n_1 \geq 10^6$ were recomputed: $0.59197\, n_1$ for $10^6 \leq n_1 \leq 2 \times 10^9$ and $0.59035\, n_1$ for $n_1 > 2 \times 10^9$, against the required $\log 1.8 \cdot n_1 = 0.58779\, n_1$. The explicit prime bound they rely on (Bennett, Martin, O'Bryant and Rechnitzer) was spot-checked for $x \leq 2 \times 10^8$, with maximum ratio 1.79628 against the stated 1.798158, and its two ranges meet at $x = 641{,}239{,}200.2$ as claimed.

The remaining external inputs are that prime bound for $x > 2 \times 10^8$, Lemma 4 of Beukers (one choice of δ gives $\Lambda \neq 0$) and Lemma 4.5 of Chudnovsky, used inside their Lemma 9.

Checks of our lemmas. The identities in Lemmas 2, 4, 6, 8 and 9 were checked by computer algebra (substituting the cycle relations and expanding both sides) or on random orbits, Lemma 4 numerically for $k_1 \leq 7$ and all odd $x < 20{,}000$, and the elimination in step 3 of the proof of Theorem 1 for every f from 33,000 to 400,000.

Proposition D.1 (digit test). Let $q \geq 17$, write $h = 3m + r$ with $r \in \{1, 2\}$, and let $\beta = 2^{r/3}$. Then $q = \lfloor 2^m \beta \rfloor$, and binary digits $m+1$ through $m + \lfloor m/2 \rfloor - 2$ of β alternate $0, 1, 0, 1, \ldots$. In particular, for $m \geq 100$ the 48 digits $m+1$ through $m+48$ alternate.

Proof. The fractional part of $2^m \beta$ is δ, made of β's digits from position $m+1$ on. Lemma 8 gives $\delta - 1/3 < 2^{k_1+1}/(3q)$ (this uses $y > x$ from Corollary 7, through $2^{k_1+k_2} \leq 2^{k_1} q$), and Lemma 5 gives

$2^{k_1} = (q+1)/x < (q+1)\sqrt{2/q}$, so

$$0 < \delta - \frac{1}{3} < \frac{2\sqrt{2}}{3}\left(1 + \frac{1}{q}\right) q^{-1/2} \le q^{-1/2} \le 2^{-\lfloor m/2 \rfloor},$$

where the middle step uses $q \ge 17$ and the last uses $q \ge 2^m$. As $1/3 = 0.010101\ldots_2$, an interval of width $2^{-\lfloor m/2 \rfloor}$ just above $1/3$ fixes all but the last two of δ's first $\lfloor m/2 \rfloor$ digits. For $m \ge 100$ that is at least 48 digits. □

Earlier versions stated this for $q \ge 71$, a threshold inherited from Proposition B.13: there the guaranteed bound on y was the smaller of $\sqrt{q/2}$ and $(3q^2/(q+1))^{1/3}$, and the cube root is the smaller one exactly when $q^2 - 70q + 1 > 0$. With Corollary 7 that crossover no longer matters. Testing only the fixed 48-digit prefix lets the scan cover every m up to $N - 48$, where N is the number of digits computed.

The run. $N = 1{,}072{,}000{,}000$ digits of each β were computed with MPFR and certified twice: by MPFR's correct rounding (the 192 guard digits are neither all 0 nor all 1, and the leading 61 digits match an exact integer cube root), and independently by the exact test $I^3 < 2^{3N+r} < (I+1)^3$ on bit lengths, where $I = \lfloor 2^N \beta \rfloor$. MPFR returned wrong values above about 2^{30} digits of precision, which these checks caught, so N was kept just below that. After XOR with `0x55`, an alternating run of 31 or more digits shows up as three equal constant bytes, so the scan is linear. The longest alternating run anywhere was 32 digits, and no position has a 48-digit alternating prefix. With the search over h for $m < 100$, this rules out every $q < 2^{1{,}071{,}999{,}953}$. Computing and scanning took about 1.5 minutes per β, and the cubing certificate about 1.5 minutes more.

E Notes on the literature and history

Literature.

- Santos's preprint proves the one- and two-step cases (Propositions 15 and 16) and conjectures that 1 is periodic only for $q = 5$ and $q = 2^e - 1$ (Conjecture 3). His Remark 5 describes three or more odd steps as far harder and likely out of reach; Theorem 1 settles three steps. The same preprint claims a proof of the Collatz conjecture, which is not accepted; only its elementary propositions are relied on here.
- Crandall (1978) and Franco and Pomerance (1995) study when orbits of $qx + 1$ maps avoid 1; Corollary B.3 matches their mechanism.
- Franco's 1990 thesis uses Diophantine approximation for the $qx + 1$ problem. Only its title was available, and it is the most important item to check for overlap.
- Steiner (1977, 1981) rules out one-block "circuits" for $q = 3$, 5, 7 with Baker's method, a related use of Diophantine approximation for cycles.
- The deep inputs are Bauer and Bennett (2002, 2018) on restricted irrationality measures, Beukers on $X^2 + D = 2^n$, and Ridout (1958) on approximation with S-unit denominators.

History of the results. The work began with the window (Lemma 8), which pins down q from h and makes a search over h far faster than a search over q. Lower bounds on x and y (Lemmas 4 to 6

and Corollary 7, preceded by the weaker Proposition B.13, which allowed $y < x$ with $y > (3q)^{1/3}$) then gave finiteness through Ridout's theorem (Proposition B.4), but no bound. Residue-class arguments settled special cases (Propositions B.5 and B.6), and the effective inequality of Bauer and Bennett gave the lower bound on k_1 (Proposition B.7) and most of the case analysis of Lemma 4 (Appendix B.5). Playing the window against the near miss for x then gave Theorem 1, once it was noticed that Bauer and Bennett's unrounded exponent lies below the golden ratio. The near-miss principle (Lemma 11) extended this to Theorems 2 and 3 and Corollary 12. The digit scan (Appendix D) was computational evidence found along the way. The final verification found and corrected a dropped constant in an intermediate step of Bauer and Bennett and confirmed their Proposition 8 for every $n_1 \geq 497$.

References

[1] M. Bauer and M. A. Bennett, Applications of the hypergeometric method to the generalized Ramanujan–Nagell equation, Ramanujan J. 6 (2002), 209–270.

[2] M. Bauer and M. A. Bennett, Ramanujan–Nagell cubics, Rocky Mountain J. Math. 48 (2018).

[3] M. A. Bennett, G. Martin, K. O'Bryant and A. Rechnitzer, Explicit bounds for primes in arithmetic progressions, Illinois J. Math. 62 (2018), 427–532.

[4] F. Beukers, On the generalized Ramanujan–Nagell equation I, Acta Arith. 38 (1980/81), 389–410.

[5] G. V. Chudnovsky, On the method of Thue–Siegel, Ann. of Math. 117 (1983), 325–382.

[6] R. E. Crandall, On the "$3x + 1$" problem, Math. Comp. 32 (1978).

[7] Z. Franco, Diophantine Approximation and the $qx + 1$ Problem, PhD thesis, UC Berkeley, 1990.

[8] Z. Franco and C. Pomerance, On a conjecture of Crandall concerning the $qx + 1$ problem, Math. Comp. 64 (1995).

[9] D. Ridout, The p-adic generalization of the Thue–Siegel–Roth theorem, Mathematika 5 (1958), 40–48.

[10] R. Santos, On the Collatz general problem $qn + 1$, arXiv:2005.00346 (preprint).

[11] R. P. Steiner, A theorem on the Syracuse problem, Proc. 7th Manitoba Conference on Numerical Mathematics (1977), 553–559.

[12] R. P. Steiner, On the "$qx + 1$ problem", q odd, Fibonacci Quart. 19 (1981), 285–288.